Core Focus on Proportions & Probability

Block 2 Review ~ Proportions and Similarity

Name _____ Period _____ Date _____

Part I – Selected Response

1. What is the value of x in the proportion $\frac{8}{10} = \frac{12}{x}$?

A. 14

B. 15

C. 16

D. 18

2. Lena bought 4 notebooks for $9.80. How much would 7 notebooks cost?

A. $12.95

B. $15.68

C. $17.15

D. $19.60

3. Marcus biked 4 miles in 18 minutes. How long would it take him to bike 10 miles at this pace?

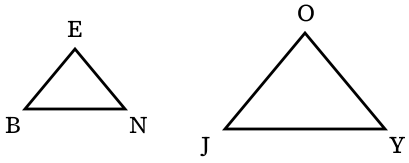
A. 40 minutes

B. 45 minutes

C. $2\frac{2}{9}$ minutes

D. 72 minutes

4. △BEN is similar to △JOY. Which side of △JOY corresponds to $\overline{BN}$ in △BEN?



A. $\overline{JO}$

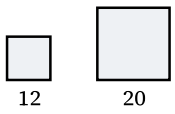
B. $\overline{OY}$

C. $\overline{YO}$

D. $\overline{JY}$

5. Which pairs of similar figures have a scale factor of 2 : 5? Circle all that apply.

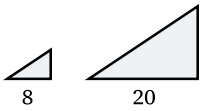
A.



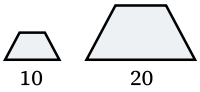
B.



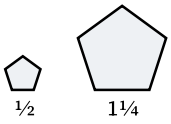
C.



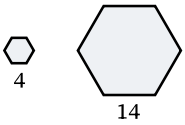
D.



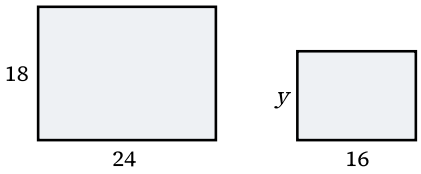
E.



F.



6. What is the value of y in the similar figures?



A. 10

B. 12

C. 14

D. 21

7. **Two similar rectangles have a scale factor of 3 : 5. For numbers 7a – 7d, determine whether each statement about the rectangles is true or false.**

7a. The ratio of their perimeters is 9 : 25. **TRUE** **FALSE**

7b. The ratio of their areas is 9 : 25. **TRUE** **FALSE**

7c. Their perimeters could be 24 cm and 40 cm. **TRUE** **FALSE**

7d. Two corresponding sides could be 6 ft and 8 ft. **TRUE** **FALSE**

Form B

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Part II – Constructed Response

Solve each proportion.

1. $\frac{5}{8} = \frac{20}{x}$

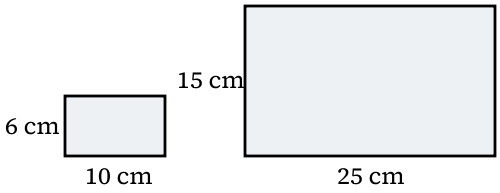
2. $\frac{12}{18} = \frac{y}{45}$

3. Priya bought 3 bags of apples for $7.47. How much would 5 bags of apples cost? Show all work necessary to justify your answer.

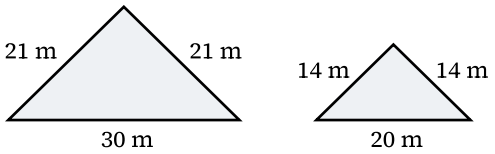
4. Carlos typed 120 words in 3 minutes. Dana typed 210 words in 5 minutes. Which person typed at a faster rate? Use mathematics to justify your answer.

Determine the scale factor for each pair of similar figures.

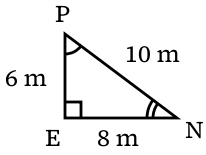
5.

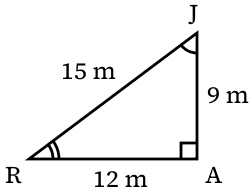


6.



7. Use the figures below to find the corresponding sides and corresponding angles to the ones identified. Matching arcs show congruent angles.





$\overline{PE}$ corresponds to _____ ∠P ≅ ∠_____

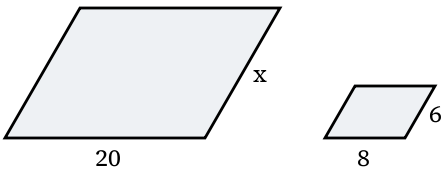
$\overline{EN}$ corresponds to _____ ∠E ≅ ∠_____

$\overline{PN}$ corresponds to _____ ∠N ≅ ∠_____

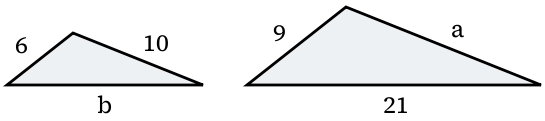
Are △PEN and △JAR congruent, similar, or neither? Explain your reasoning.

Each pair of shapes below are similar. Solve for each variable. Show all work necessary to justify your answer.

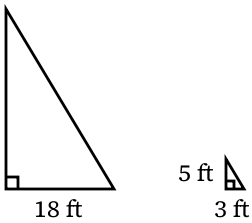
8.



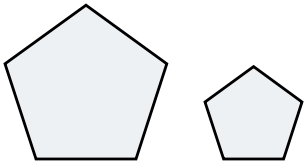
9.



10. Nora wants to know the height of the flagpole at her school. One afternoon, the flagpole's shadow along the ground is 18 feet long. At the same time, Nora's own shadow is 3 feet long. Nora is 5 feet tall. Find the height of the flagpole. Use mathematics to justify your answer.



11. Use the similar regular pentagons below.



a. Find their scale factor (large to small).

b. Find the ratio of their perimeters.

c. Find the ratio of their areas.

d. Suppose the larger pentagon has a perimeter of 100 cm. What is the perimeter of the smaller pentagon? Show all work necessary to justify your answer.

12. Two similar rectangles have perimeters of 12 feet and 48 feet. The smaller rectangle has an area of 5 ft^2. Marco used this information to find the area of the larger rectangle. Unfortunately, he made an error in his work. Explain the error and then find the correct area of the larger rectangle.

Scale factor is 12 : 48, which is 1 : 4

Ratio of their areas is 1 : 4

Let x = the area of the larger rectangle

$\frac{1}{4} = \frac{5}{x}$

$x = 20$

The area of the larger rectangle is 20 ft^2.

13. A map has a scale of 3 centimeters : 40 kilometers. On the map, the distance from Owen's town to the lake is 7.5 centimeters. How many kilometers is the lake from Owen's town? Show all work necessary to justify your answer.

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Part III – Problem Solving

Jada stands next to a light pole in a parking lot. She is 5 feet tall and her shadow along the ground has a length of 3 feet. At the same time, the light pole casts a shadow along the ground that has a length of 12 feet. What would the length of the light pole's shadow along the ground be if the pole were 5 feet shorter?

